Abbilash Shivaparan

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Software and data-focused Information Science student with experience shipping full-stack web and desktop apps using Next.js, PostgreSQL, and FastAPI, automating data workflows in Python and SQL at NJ TRANSIT, and building RAG pipelines and evaluating LLMs at Pfizer and in UMD research.

EXPERIENCE

NJ TRANSIT • Data Analytics – IT Intern • 05/2024 – 09/2025

- Built SQL queries and Excel reports generating weekly operational statistics for 5,000+ ride records; automated geospatial validation of 2,000+ addresses using Haversine calculations, reducing manual review by 60%.
- Developed Python scripts using Google Reverse Geocoding API to rank top 25 high-frequency pickup locations for route optimization.
- Built 3 Power BI dashboards visualizing ridership trends and operational KPIs; created Crystal Reports dashboard flagging 150+ mismatched location IDs and presented enhancement roadmap to management.

Pfizer (via Extern) • AI-Powered Document Insights Extern • 06/2026 – 08/2026

- Built a RAG pipeline over pharmaceutical certificate documents; resolved corrupted PDF text extraction by switching to PyMuPDF and increased retrieval depth from 3 to 10 chunks to recover multi-section answers.
- Applied prompt engineering and LLM integration to pharmaceutical knowledge-management use cases.

Windsor Internal Medicine and Geriatrics • Office Assistant – IT Support • 06/2022 – Present

- Implemented access controls and security protocols for patient data protection; performed Windows patching across 10+ workstations maintaining HIPAA compliance.

RESEARCH

University of Maryland, Chan Lab • Undergraduate Researcher • 09/2026 – Present

- Evaluating how accurately frontier and local LLMs extract claims and evidence from research papers compared to human review, using 205 model-drafted discourse graph nodes across 27 papers.
- Leading local model setup (Gemma/Llama) and scoring with precision, recall, and F1; categorizing model errors by type, such as missing context, reframed meaning, and quote mismatches.

PROJECTS

DataProcessBot – RAG-powered desktop app for document Q&A (Tauri, Python FastAPI, ChromaDB)

- Built a cross-platform desktop application enabling users to query their own documents through a retrieval-augmented generation pipeline powered by ChromaDB.
- Architected a Python FastAPI backend paired with a Tauri front end, delivering a lightweight, native-feeling desktop experience for document search and Q&A.
- Implemented grounded Q&A with relevance-threshold retrieval and source citations, plus auto-summarize, extract, and flag actions that run automatically on every upload.

SubleaseHer – Full-stack sublease marketplace (Next.js, PostgreSQL, Prisma) | subleaseher.vercel.app

- Designed and deployed marketplace connecting students with housing; engineered PostgreSQL schema with Prisma ORM, integrated Cloudinary for image delivery, and implemented NextAuth.js authentication.
- Built listing search with keyword, location, price, bed, and date filters, listing-based messaging with unread counts, favorites, and 1–5 star user reviews.

NBA Player DNA – ML clustering dashboard (Python, scikit-learn, Streamlit) | nba-player-dna.streamlit.app

- Implemented unsupervised pipeline: PCA for dimensionality reduction + K-Means clustering of player statistics, segmenting 500+ players into 8 archetypes.
- Built a Team Fit Analyzer scoring free-agent fit (55% archetype gap, 25% PIE, 10% points, 10% assists) within a 5-page dashboard that includes player comparison and shot charts.

Electric Vehicle Registration Analysis – SQL, 3NF database design, ER diagrams

- Designed a normalized (3NF) relational database for 100K+ WA State EV records; found the top 3 manufacturers account for 78% of registrations and identified the highest-adoption counties.

EDUCATION

Bachelor of Science in Information Science, Minor in Data Science

University of Maryland • College Park, MD • Expected May 2027

- Activities: Cyber Security Club, AWS Practitioner Club, UMD Club Basketball

SKILLS

Data & BI: SQL, Excel (PivotTables, XLOOKUP), Power BI, Crystal Reports
Programming: Python, R, Java, JavaScript, React, Next.js, TypeScript
Databases & Web: PostgreSQL, Prisma ORM, NextAuth.js, Tailwind CSS, Cloudinary, FastAPI, Tauri
ML & Data Science: scikit-learn, pandas, PCA, K-Means, statistical analysis, Streamlit, RAG, LLMs, LlamaIndex, ChromaDB, embeddings, NLP
Tools: Git, GitHub, Oracle SQL Developer, Figma, Google APIs, Vercel